FRAUD DETECTION WITH KNIME

KNIME Student Challange

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Problem Overview

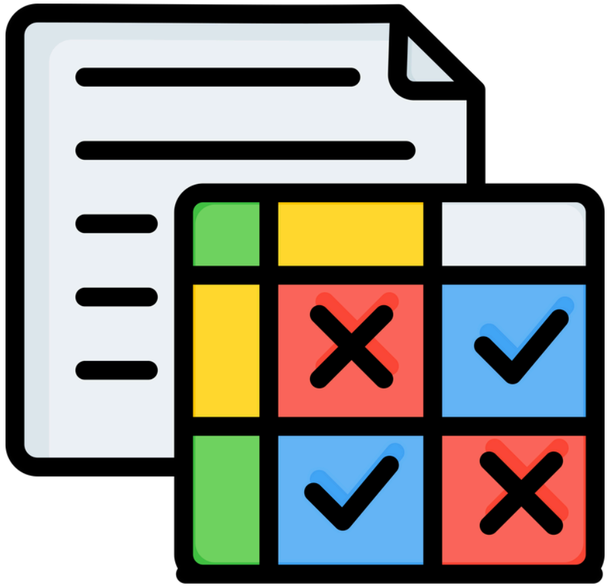
The exponential growth of e-commerce platforms has led to an increase in fraudulent activities, posing serious financial and reputational risks to businesses. Fraudsters often exploit vulnerabilities in payment systems, user authentication, and transaction monitoring.

Impact:

- Financial Losses
- Loss of Trust and Reputation
- Operational and Compliance Costs

Methodology

- We adopted the CRISP-DM methodology to guide the structured development of our fraud detection workflow.
- The analytical techniques include statistical outlier detection using interquartile ranges, Decision Trees, Random Forests, and an Autoencoder neural network.
- The dataset was split into training and testing sets to ensure fair model evaluation.



QUARTILLES

DECISION TREES

RANDOM FOREST

LOGISTIC REGRESSION

AUTOENCODER

Dataset

To achieve the defined objectives, a dataset comprising 1,472,952 transactions performed in the year 2024 was used, containing 16 variables that provide detailed information about each transaction.

TARGET VARIABLE "IS FRAUD"

This allows the use of supervised learning models and enables objective performance evaluation using classification metrics.

Variables:

- Transaction ID
- Customer ID
- Transaction Amount
- Transaction Date
- Payment Method
- Product Category
- Quantity
- Customer Age
- Customer Location
- Device Used
- IP Address
- Shipping Address
- Billing Address
- Is Fraudulent
- Account Age Days
- Transaction Hour

Challenges

- **IMBALANCED DATASET**

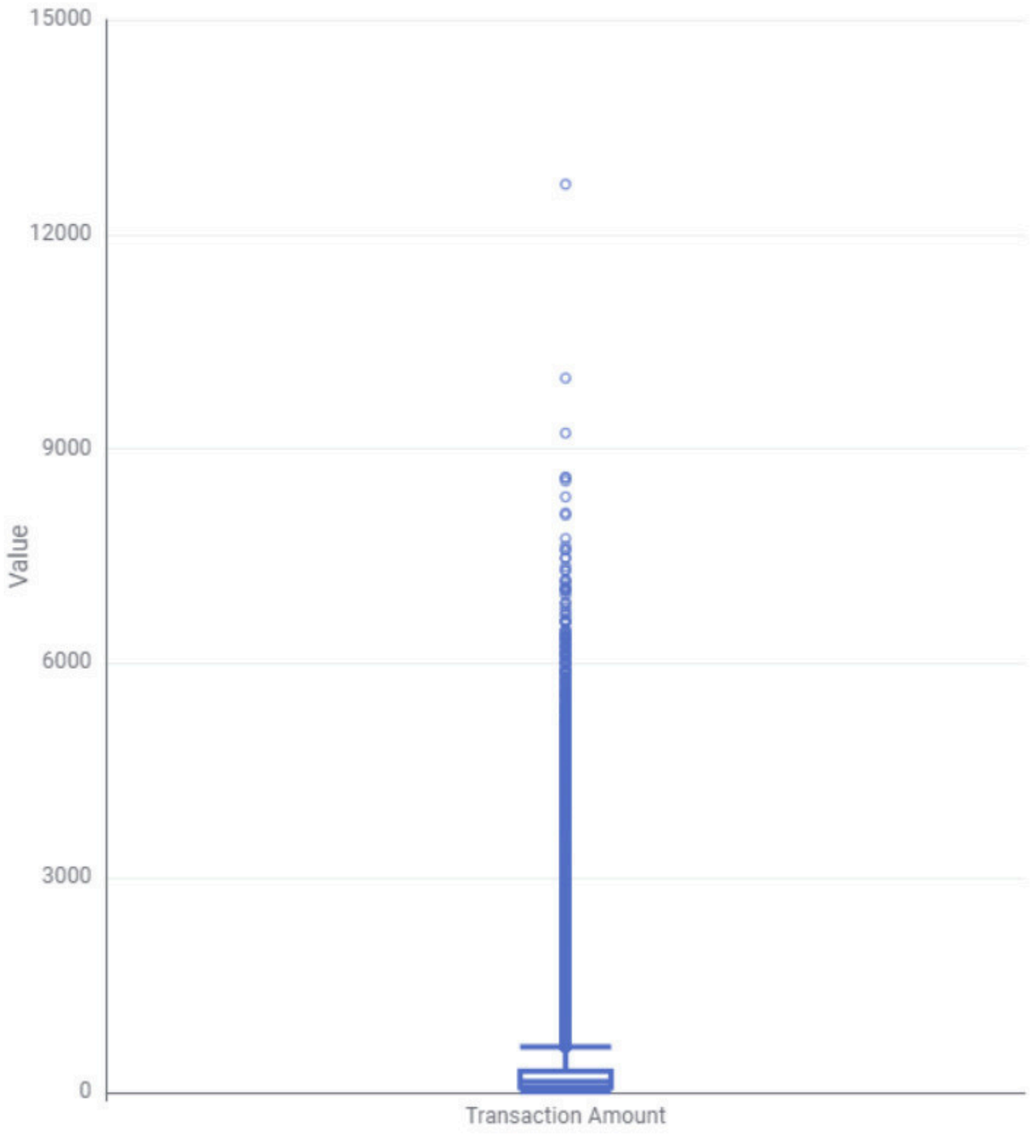
Out of the 1,472,952 transactions available in the dataset, only 73,838 were labeled as fraudulent, corresponding to approximately 5.28% of the total.

- **FALSE POSITIVES VS. FALSE NEGATIVES**

It's difficult to balance catching real fraud without overwhelming systems with false alarms.

- **OUTLIERS COMPLEXITY**

Distinguishing between legitimate anomalies and real fraud is difficult and can lead to misclassification.

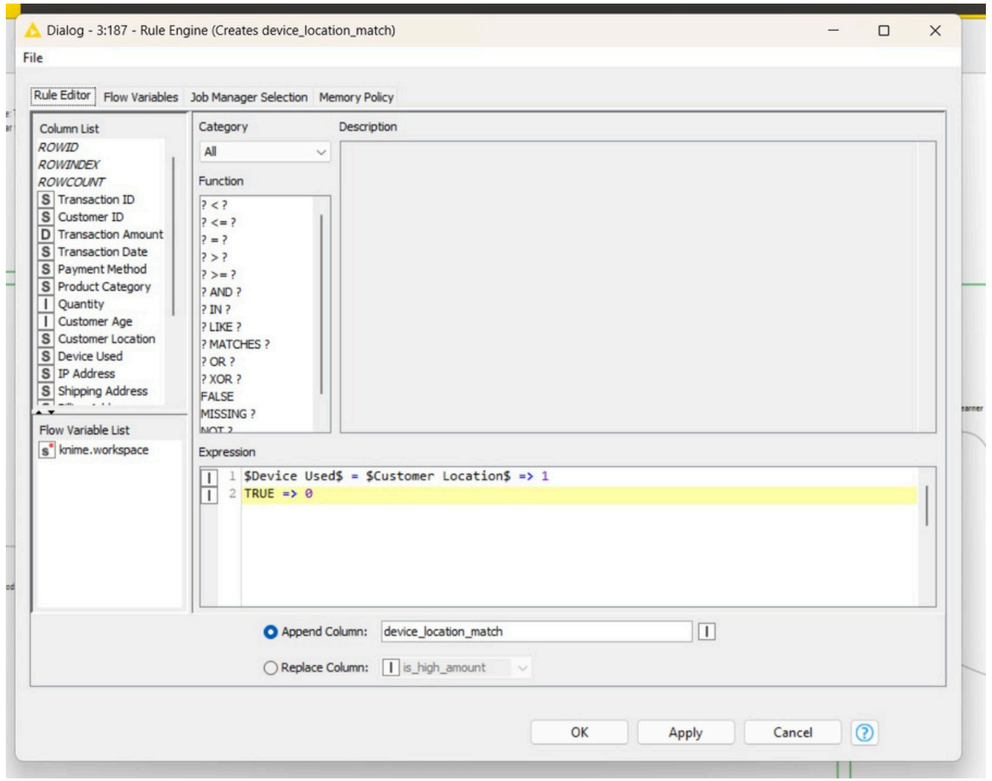


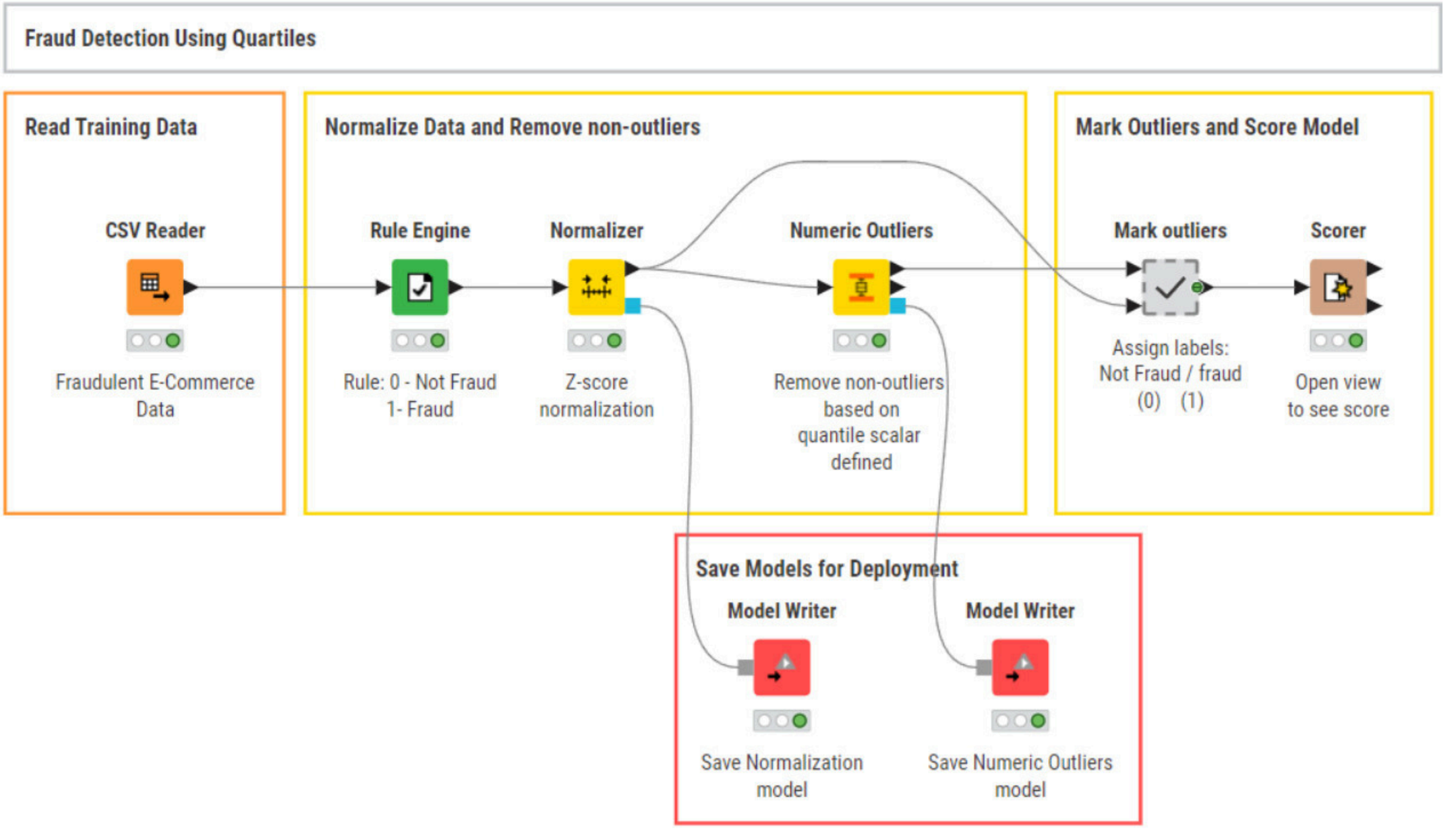
Data Preparation

- The dataset had no missing values, so no imputation was required.
- Outliers were preserved, as they may represent suspicious behavior important for fraud detection.
- Negative ages were kept intentionally, considered potential indicators of fraud or data manipulate;
- Z-score normalization was applied to numerical variables for models sensitive to scale, such as Quartilles, Logistic Regression
- Unnecessary columns such as transaction and customer IDs were removed from model training to avoid introducing bias or noise.
- Each model used a tailored preprocessing pipeline, ensuring fair evaluation and optimal performance.

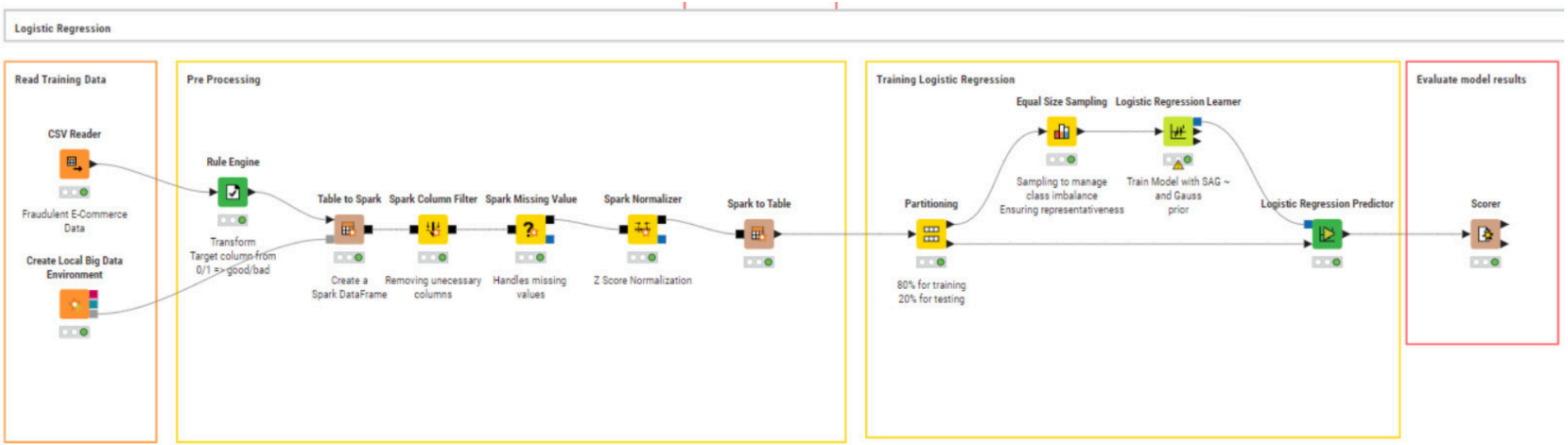
Feature Enginering

- **day_period:** a categorical variable derived from the transaction hour, indicating the time of day (e.g., Morning, Afternoon, Night).
- **amount_per_hour:** a numerical ratio between the transaction amount and the transaction hour (plus one), aiming to capture unusual transaction values at atypical hours.
- **device_location_match:** a binary indicator showing whether the device used in the transaction matches the customer's registered location. This rule assigns a value of 1 if the device used matches the customer's location, and 0 otherwise, aiding in anomaly detection for potential fraud cases.

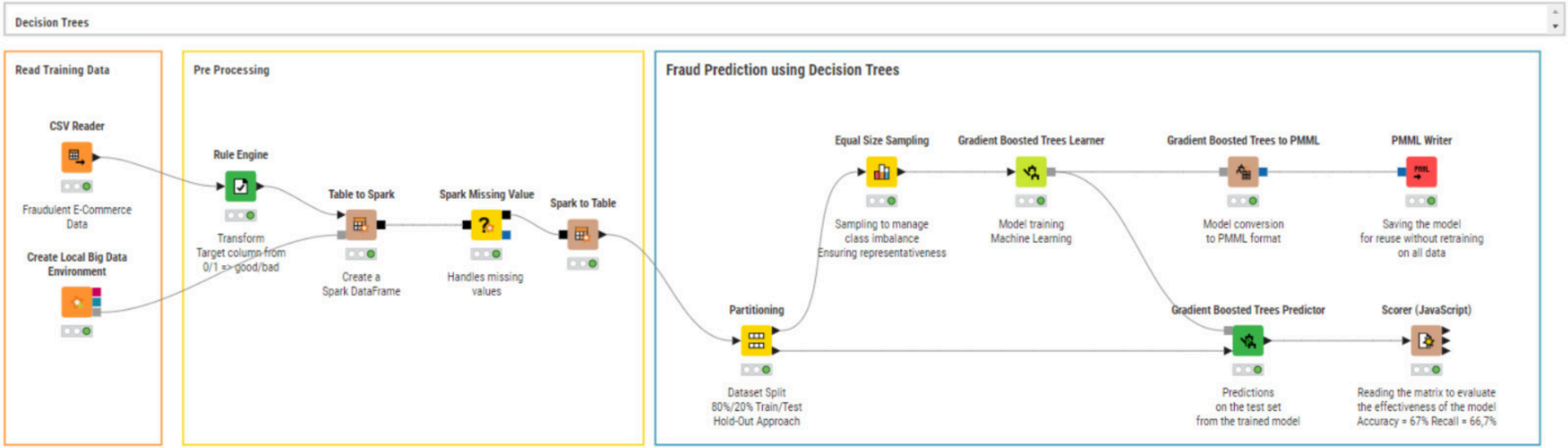



Fraud Detection Using Quartiles
Read Training Data
CSV Reader
Fraudulent E-Commerce Data
Normalize Data and Remove non-outliers
Rule Engine
Rule: 0 - Not Fraud
1- Fraud
Normalizer
Z-score normalization
Numeric Outliers
Remove non-outliers based on quantile scalar defined
Mark Outliers and Score Model
Mark outliers
Assign labels:
Not Fraud / fraud
(0) (1)
Scorer
Open view to see score
Save Models for Deployment
Model Writer
Save Normalization model
Model Writer
Save Numeric Outliers model

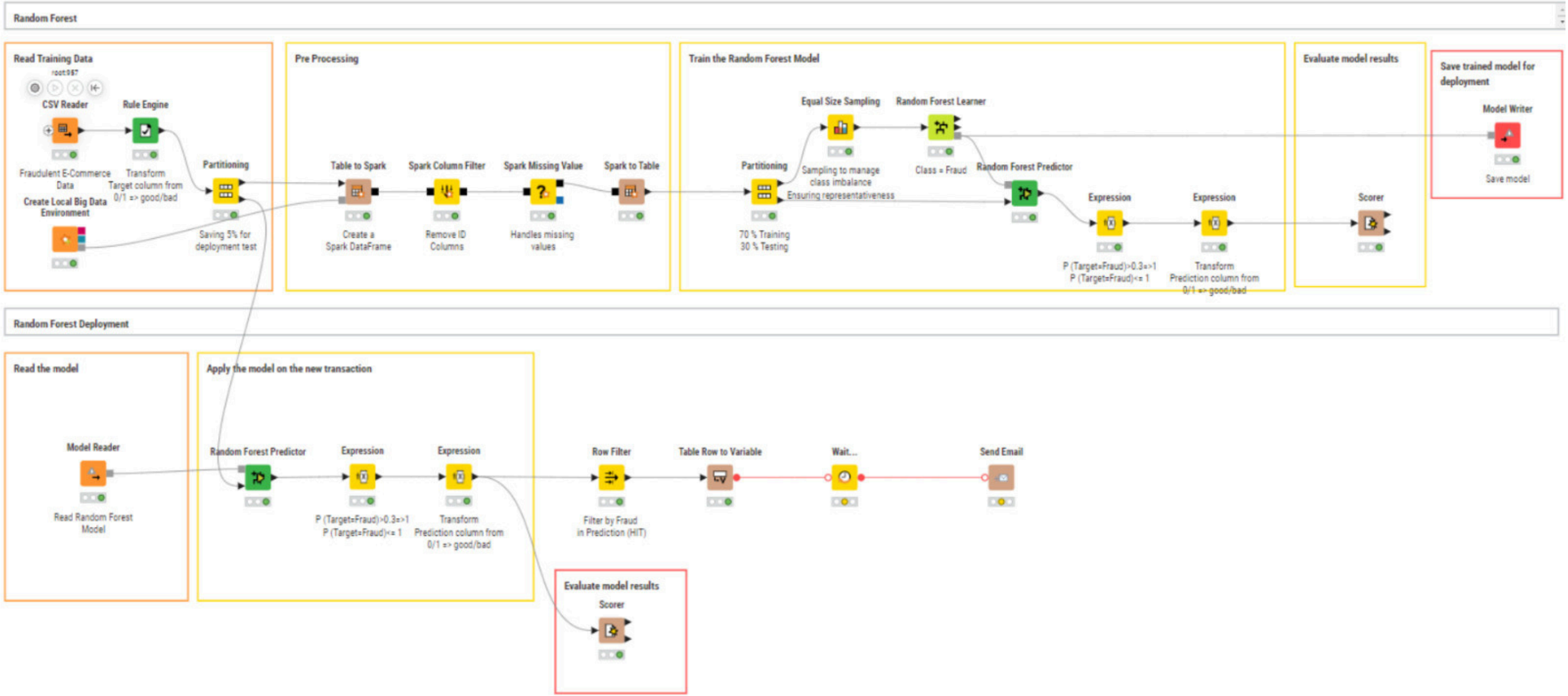
Logistic Regression



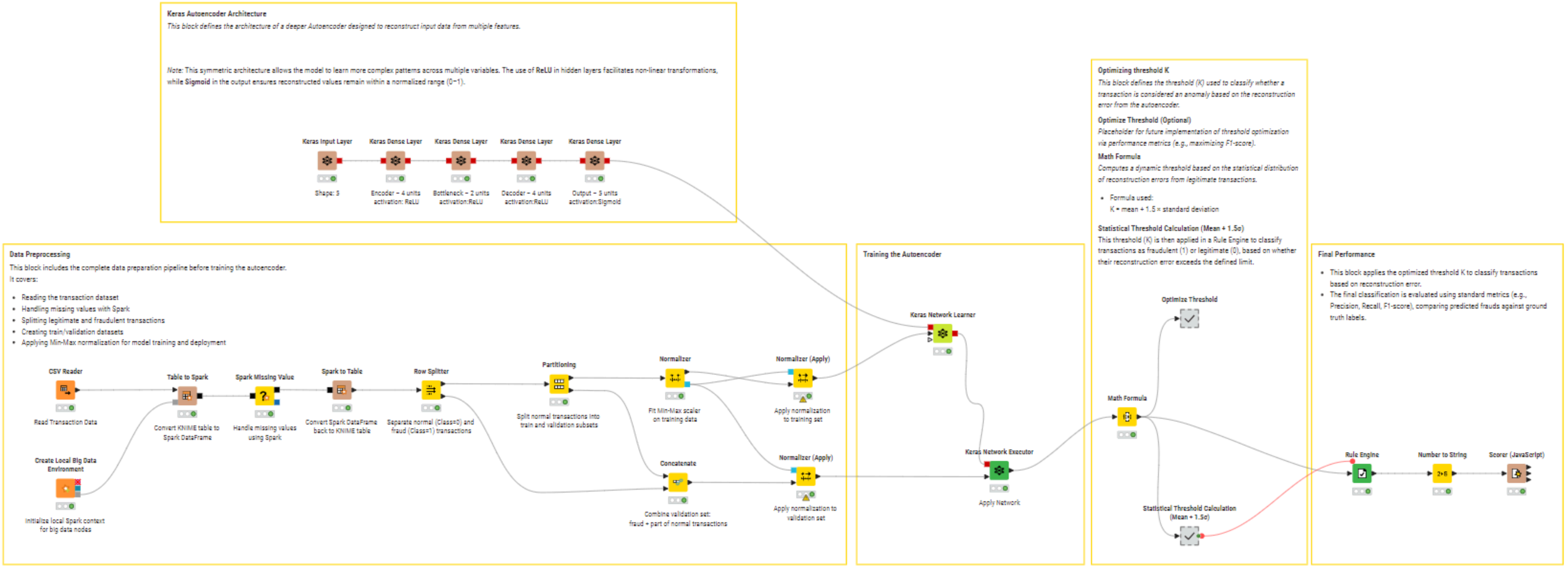
Decision Trees



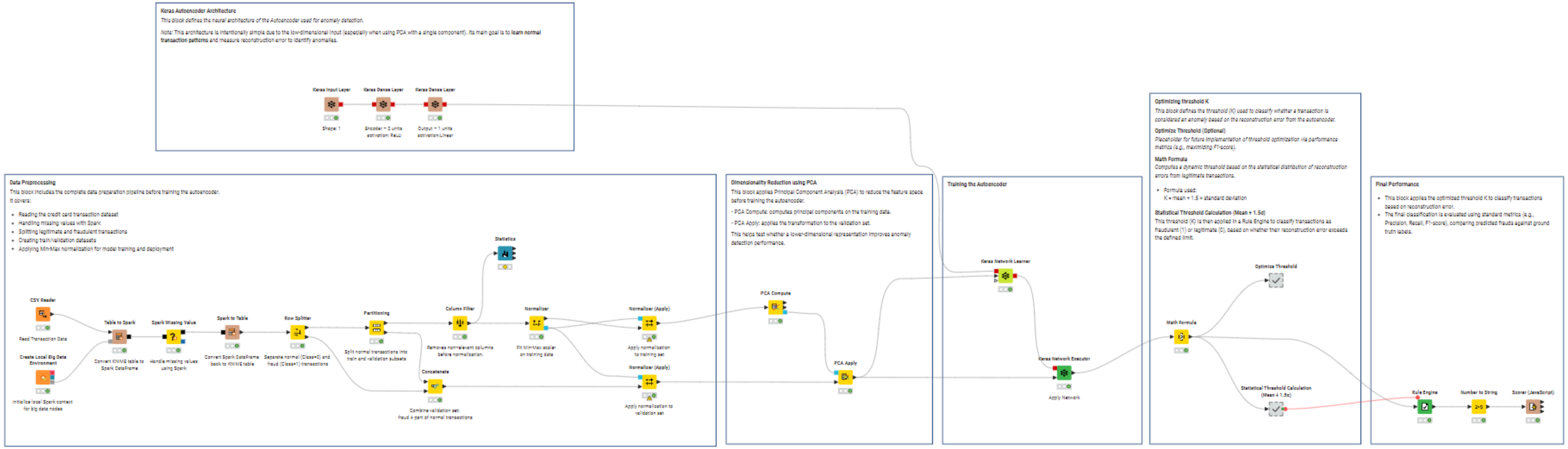
Random Forest + Deployment



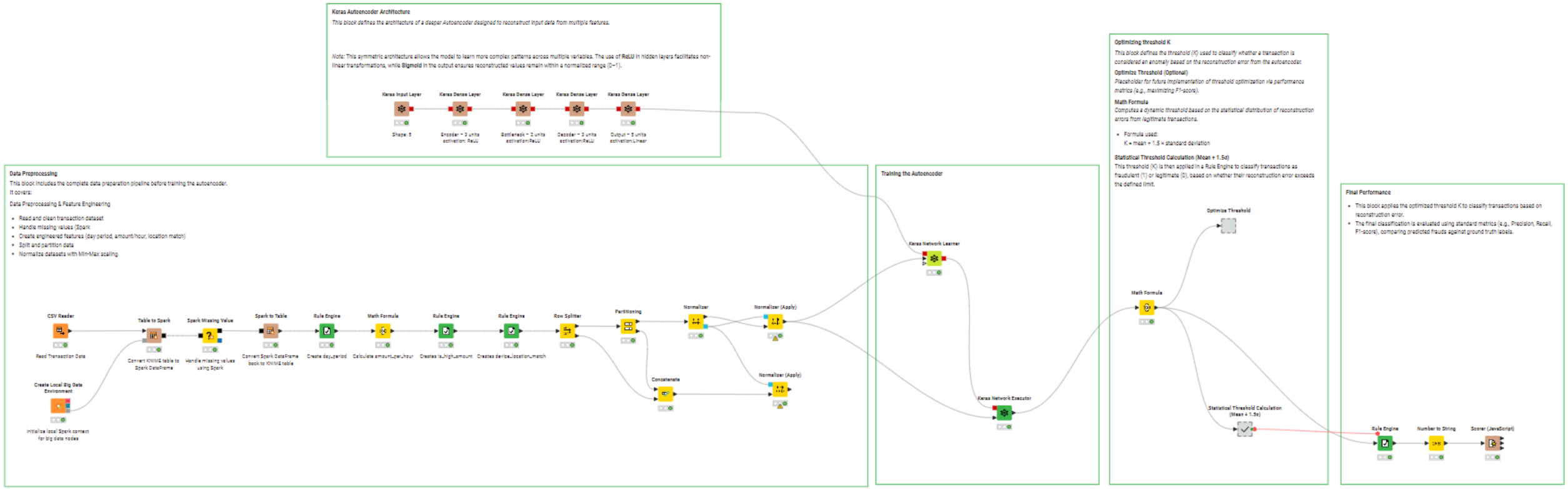
Autoencoder



Autoencoder + PCA



Autoencoder + FE



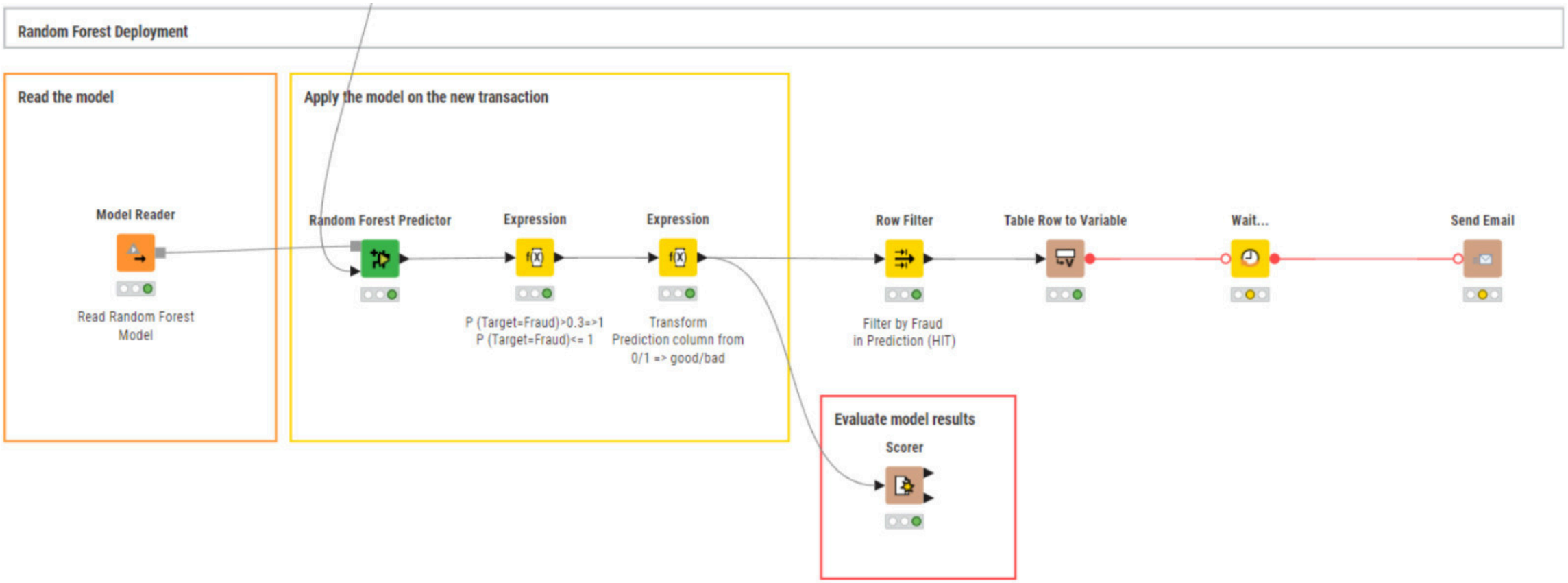
- Decision Trees and Autoencoder + PCA stood out for their high recall, making them effective first-line fraud detectors.
- Quartile method achieved the best overall balance, with the highest accuracy (93.63%) and good interpretability.
- Random Forest and Logistic Regression showed lower overall performance due to poor precision, limiting their practical use.

Results

Models	Quartilles	Decision Trees	Random Forest	Logistic Regression	Autoencoder
Accuracy	93.63%	64,00%	43,50%	60,70%	44,00%
Precision	30.80%	10,13%	6,67%	5,90%	13,40%
Recall	21.45%	88,89%	88,90%	53,90%	79,90%
F1-Score	25.25%	18,05%	12,40%	10,60%	23,00%

Deployment

- Deployed the Random Forest model to automatically scan incoming transactions.
- Used a Row Filter to isolate predicted fraud cases.
- For each fraud hit, an automated email alert is sent with full transaction details.
- Enabled near real-time fraud monitoring with KNIME's built-in automation tools.

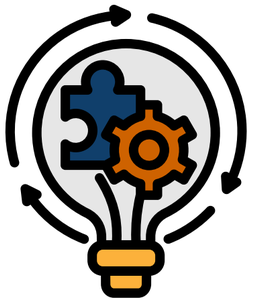


Conclusions & Impact

- No single superior model: Each involves trade-offs; selection depends on system goals and constraints.
- E-commerce recommendation: Use a hybrid model combining performance, speed, and interoperability.
- We suggest a 3-layer system:
    1. Autoencoder + PCA: Flags anomalies early
    2. Decision Tree: Main classifier
    3. Quartile model: Simple, accurate validator
- Outcome: Broad coverage, real-time triage, and explainability — ideal for scalable, robust fraud detection.

Future Work

- **Literature Review:** To support preprocessing decisions (e.g., handling of outliers and anomalies).
- **Feature Engineering:** Create derived variables (e.g., transaction frequency, time gaps, value ratios) to capture fraud patterns.
- **Validation Strategy:** Apply stratified and temporal cross-validation for more robust and realistic model evaluation.
- **Model Explainability:** Use SHAP and LIME to interpret predictions, ensuring transparency and ethical deployment in fraud detection.



THANK YOU!